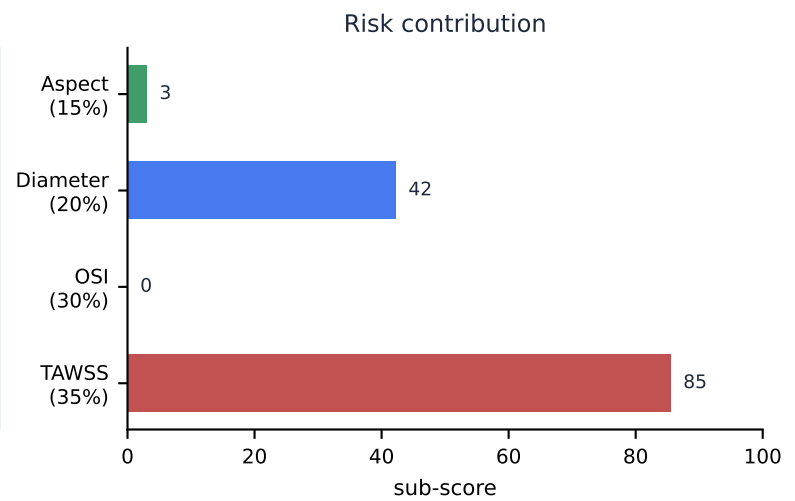
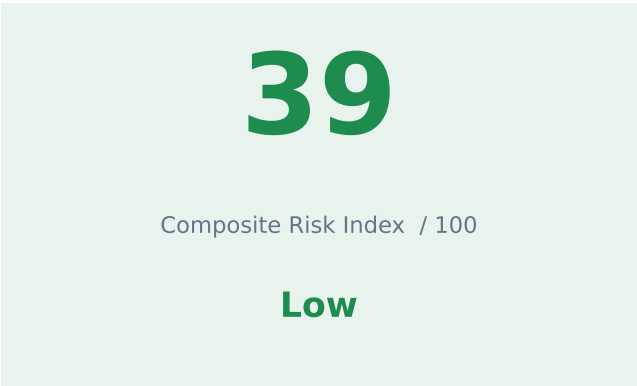


NeuroFlow CFD Analyst

Clinical Hemodynamic Assessment

Case PT-2026-0103 · generated 2026-08-05 11:27



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	0.346 Pa	< 0.4 Pa	FLAGGED
TAWSS — parent artery	2.939 Pa	reference	normal
OSI — aneurysm dome	0.008	> 0.2	normal
RRT — aneurysm dome	7.17 Pa ⁻¹	> 3.0	FLAGGED
ECAP — aneurysm dome	0.173 Pa ⁻¹	> 1.0	normal
NWSS (sac / parent)	0.118	—	normal
LSAR (relative, <10% parent)	71.3 %	—	normal
LSAR (absolute, <0.4 Pa)	78.3 %	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
5.38 mm	4.83 mm	0.75	1.11
Volume	Surface area	Non-sphericity	
63.4 mm ³	61.1 mm ²	0.001	

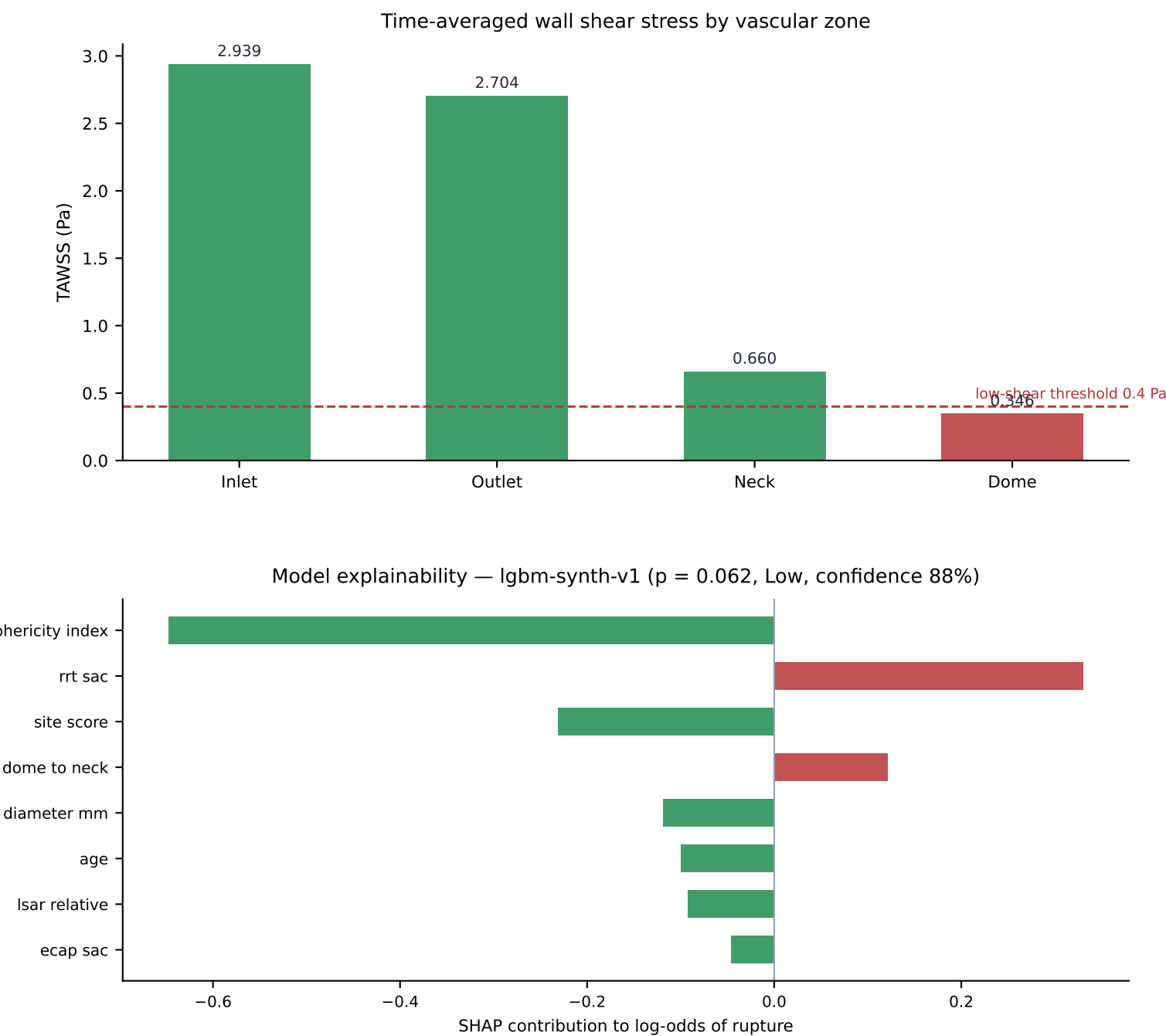
Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 238994 · SIMPLE solution converged in 185 iterations
WSS converted from OpenFOAM kinematic units (m²/s²) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution $\tau = 4 \cdot \mu \cdot Q / (\pi \cdot r^3)$.

NeuroFlow CFD Analyst

Zone Analysis and Model Explainability

Case PT-2026-0103



Assessment

Computational fluid dynamics analysis of case PT-2026-0103. Time-averaged wall shear stress at the aneurysm dome measured 0.346 Pa against 2.939 Pa in the parent artery (normalised WSS 0.118), with an oscillatory shear index of 0.008. 71.3% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 7.17 Pa⁻¹. Sac morphology measured from the reconstructed surface: maximum diameter 5.38 mm, neck 4.83 mm, aspect ratio 0.75, volume 63.4 mm³. Composite Risk Index 39/100 (Low).